

Matrix functions and stability

Optimal sign approximation with a multiplication budget

Difficulty: challenging **Importance:** interesting to the community

Status: open; explicit 2026 problem

Problem statement

For $m \in \mathbb{N}_0$ and $0 < \delta < 1$, put

$$I_\delta = [-1, -\delta] \cup [\delta, 1].$$

Let $\mathcal{P}_m$ consist of real polynomials computed from $1, x$ by straight-line programs using at most m non-scalar multiplications; real linear combinations cost nothing. Define

$$E_m(\delta) = \inf_{p \in \mathcal{P}_m} \max_{x \in I_\delta} |p(x) - \text{sign}(x)|.$$

Determine matching asymptotic upper and lower bounds for $E_m(\delta)$, with the dependence on both m and δ explicit. Infima avoid assuming attainment. The arithmetic model concerns a single polynomial identity valid for matrices of every size.

References and status evidence

Amsel et al., [Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop](#), §6.3, Problem 6.3 (2026). Rubensson, Jarlebring and Lorentzon, [Recursive expansion of the matrix step function using polynomials of degree eight](#), §7 (June 2026), still identifies optimal recursive expansion as open. Its particular algorithm does not establish the unrestricted optimum above.